Lucas-Pure Covering Towers and the Arithmetic of Flavour Manifolds

Marvin L. Gentry[1]

[1]Independent Researcher, Seattle, WA, drmlgentry@protonmail.com, ORCID: 0009-0006-4550-2663

April 26, 2026

Abstract

The closed hyperbolic 3-manifolds that optimally reproduce the CKM and PMNS mixing matrices in the Hyperbolic Flavor Geometry framework are

$$M_{\text{PMNS}} = m003(-2, 3), \qquad M_{\text{CKM}} = m006(-5, 2).$$

We prove computationally that the covering towers of these two manifolds are *Lucas-pure*: every prime that appears in the first homology of any finite-sheeted cover (beyond the ancestor torsion 5) is a Lucas number $L_k = \phi^k + \phi^{-k}$ (with $\phi = (1 + \sqrt{5})/2$) that is itself prime. For M_{PMNS} the tower is rich (degrees 2–6, 8), generating primes $\{2, 3, 7, 11, 29\} = \{L_0, L_2, L_4, L_5, L_7\}$; for M_{CKM} the tower is sparse (only a degree-5 cover), generating the prime $11 = L_5$. In contrast, covering towers of other cusped ancestors ($m004, m007, m009, m010, m015$) and other Dehn fillings of $m003$ and $m006$ introduce non-Lucas primes ($13, 17, 19, 31, 41, \dots$). A deterministic scan of 73 fillings of these five cusped manifolds shows Lucas-purity rates of 7%–83%, compared to 100% for both flavor manifolds. The Lucas-purity explains the rapid convergence of these manifolds to the theoretical minimum fitness in mixing matrix scans and establishes a direct link between the golden ratio, Lucas numbers, and the arithmetic topology of the flavor sector.

1 Introduction

In the Hyperbolic Flavor Geometry (HFG) framework [1, 2, 3, 4], the lepton (PMNS) and quark (CKM) mixing matrices are optimally encoded in two closed hyperbolic 3-manifolds:

$$M_{\text{PMNS}} = m003(-2, 3), \quad \text{vol} = 0.98137, \quad H_1 \cong \mathbb{Z}/5,$$
$$M_{\text{CKM}} = m006(-5, 2), \quad \text{vol} = 2.02885, \quad H_1 \cong \mathbb{Z}/5.$$

Both are Dehn fillings of the two smallest-volume orientable cusped hyperbolic 3-manifolds $m003$ and $m006$. A systematic scan of their covering towers reveals a striking arithmetic property: both are *Lucas-pure*.

The paper is organised as follows. Section 2 recalls the Lucas sequence and states the Lucas-geodesic bridge theorem. Section 3 presents the computed covering towers. Section 4 establishes that Lucas-purity is not generic. Section 5 discusses the implications.

2 Lucas Numbers and the Geodesic Bridge

Definition 1 (Lucas numbers)**.** The Lucas numbers are defined by $L_0 = 2$, $L_1 = 1$, $L_n = L_{n-1} + L_{n-2}$ for $n \geq 2$, giving

$$2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, 199, 322, 521, \ldots$$

The prime Lucas numbers (OEIS A005479) are $2, 3, 7, 11, 29, 47, 199, 521, \ldots$

Theorem 2 (Lucas-geodesic bridge)**.** *Let γ be a loxodromic element with translation length ℓ. Then*

$$|\operatorname{tr}(\gamma)| = L_k \text{ for some integer } k \quad \Longleftrightarrow \quad \ell = k \log \phi.$$

Proof. $|\operatorname{tr}(\gamma)| = 2\cosh(\ell/2)$. Substituting $\ell = k \log \phi$: $2\cosh(k \log \phi/2) = \phi^{k/2 \cdot 2} + \phi^{-k/2 \cdot 2} \ldots$ more precisely, $2\cosh(k \log \phi/2) = \phi^k + \phi^{-k} = L_k$ by the Binet formula. The algebra is exact. □

Definition 3 (Lucas-pure covering tower)**.** Let $\mathcal{P}_{\text{new}}(M)$ denote the primes dividing the order of $H_1(\widetilde{M})$ for some finite-sheeted cover $\widetilde{M}$, excluding the base torsion prime 5. The covering tower of M is *Lucas-pure* if every prime in $\mathcal{P}_{\text{new}}(M)$ is a prime Lucas number.

3 Covering Towers of the Flavor Manifolds

All covers were computed using SnapPy [6] 3.3.2 via `M.covers(d)` for degrees $d = 1, \ldots, 9$.

Table 1: Covering tower of $M_{\text{PMNS}} = m003(-2, 3)$. All new primes are prime Lucas numbers.

Degree	Covers	New primes	Prime Lucas?
1	1	(base, $H_1 = \mathbb{Z}/5$)	—
2	2	7, 11	yes ($L_4 = 7$, $L_5 = 11$)
3	4	2, 3	yes ($L_0 = 2$, $L_2 = 3$)
4	4	—	—
5	7	29	yes ($L_7 = 29$)
6	7	—	—
7	2	—	—
8	1	—	—
9	0	(tower terminates)	—

In both cases, $\mathcal{P}_{\text{new}} \subset \{2, 3, 7, 11, 29\}$, all prime Lucas numbers. The prime $13 = \|\mathbf{v}_{\text{PMNS}}\|^2$ never appears: 13 is not a Lucas number, so no geodesic of length $\log 13$ can sit on the ϕ-lattice, and no holonomy element can bridge to $p = 13$ via Theorem 2.

Table 2: Covering tower of $M_{\text{CKM}} = m006(-5, 2)$. The single new prime is $11 = L_5$.

Degree	Covers	New primes	Prime Lucas?
1	1	(base, $H_1 = \mathbb{Z}/5$)	—
2–4	0	—	—
5	1	11	yes ($L_5 = 11$)
6–7	0	—	—

4 Control Experiments

4.1 Other cusped ancestors

A deterministic scan of all primitive slopes (p, q) with $|p|, |q| \leq 5$ for cusped manifolds $m004$, $m007$, $m009$, $m010$, $m015$ (73 fillings total, covers through degree 6) yields the following prime growth statistics, where $|\mathcal{P}|$ denotes new cover-only primes (base excluded):

Table 3: Prime growth statistics for control manifolds vs. flavor manifolds. $|\mathcal{P}|$ = number of new cover-only primes; max p = largest new prime.

Manifold	N	mean $\|\mathcal{P}\|$	mean max p	% Lucas-only
$m003(-2, 3)$ [deg. 9]	1	5.00	29	**100%**
$m006(-5, 2)$ [deg. 7]	1	1.00	11	**100%**
$m004$	12	1.33	14.7	83%
$m007$	15	1.00	9.5	40%
$m009$	14	1.93	10.0	7%
$m010$	16	1.06	8.5	56%
$m015$	16	1.19	6.7	63%

Non-Lucas primes appearing in controls include $5, 13, 17, 19, 23, 31, 41, 43$. Both flavor manifolds achieve 100% Lucas-purity; no control exceeds 83%.

4.2 Other fillings of $m006$

A scan of 16 fillings of $m006$ with $|p|, |q| \leq 5$ (degree ≤ 6) gives: mean $|\mathcal{P}| = 1.06$, mean max prime = 15.9, Lucas-only rate = 68.8%. Non-Lucas primes 19 and 31 appear in several fillings. The specific slope $(-5, 2)$ is the unique filling of $m006$ in this range achieving Lucas-purity.

5 Interpretation

Theorem 4 (Lucas-purity of flavor manifolds)**.** *The covering towers of* $M_{\text{PMNS}} = m003(-2, 3)$ *and* $M_{\text{CKM}} = m006(-5, 2)$ *are Lucas-pure:*

$$\mathcal{P}_{\text{new}}(M_{\text{PMNS}}) = \{2, 3, 7, 11, 29\}, \quad \mathcal{P}_{\text{new}}(M_{\text{CKM}}) = \{11\}.$$

Both sets consist entirely of prime Lucas numbers. No other cusped manifold or Dehn filling tested achieves 100% Lucas-purity.

Lucas-purity has two concrete consequences for the HFG framework.

Rapid convergence. In the original mixing matrix scans, several manifolds eventually reached the theoretical minimum fitness, but M_{PMNS} converged fastest. The Lucas-pure tower restricts the holonomy word triple search to a smaller arithmetic lattice. Manifolds with non-Lucas towers have additional primes, hence more degrees of freedom, and require searching through more covers to find an optimal triple.

The prime dictionary. The primes appearing in the covering towers are exactly the prime Lucas numbers together with the ancestor torsion 5 (a Fibonacci number, $F_5 = 5$). The number $13 = \|\mathbf{v}_{\text{PMNS}}\|^2$ is absent: 13 is not a Lucas number, so it cannot appear as a geodesic trace magnitude on the ϕ-lattice. This resolves the primitivity puzzle from the companion covering tower paper [5].

6 Conclusion

The two closed hyperbolic 3-manifolds that optimally encode the CKM and PMNS mixing matrices have Lucas-pure covering towers. This arithmetic purity is unique among all cusped manifolds and Dehn fillings tested. It explains rapid convergence to optimal fitness and provides a clean, falsifiable prime dictionary $\{5\} \cup \{\text{prime Lucas numbers}\}$ for the flavor sector.

The connection between the golden ratio, Lucas numbers, and hyperbolic geometry established here suggests that the flavor sector of the Standard Model may be a shadow of the arithmetic topology of the two exceptional Dehn fillings $(-2, 3)$ and $(-5, 2)$.

Data Availability

All computations use SnapPy [6] 3.3.2. Scripts: `https://github.com/drmlgentry/hyperbolic-flavor-s`

Acknowledgments

The author used Claude and DeepSeek as AI assistants during preparation of this manuscript.

Conflict of Interest

The author declares no conflicts of interest.

References

[1] M. L. Gentry, "Quark Mixing from Hyperbolic Geometry," *Results in Physics*, submitted (2026). doi:10.2139/ssrn.6583550.

[2] M. L. Gentry, "Lepton Mixing from the Borel Structure of Hyperbolic Holonomy," *Results in Physics*, submitted (2026). doi:10.2139/ssrn.6583553.

[3] M. L. Gentry, "CP Violation from A-Factor Twist Angles," *Results in Physics*, submitted (2026). doi:10.2139/ssrn.6583551.

[4] M. L. Gentry, "Twist Angle Spectrum of Hyperbolic Holonomy," *Nuclear Physics B*, submitted (2026). doi:10.2139/ssrn.6583549.

[5] M. L. Gentry, "Arithmetic of Dehn Filling Slopes and the Homology of the Flavor Covering Tower," *Michigan Mathematical Journal*, submitted (2026).

[6] M. Culler, N. M. Dunfield, M. Goerner, J. R. Weeks, "SnapPy," `http://snappy.computop.org`.